

Miraj Chamara Samarakkody

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RESEARCH INTERESTS

Manifold learning and geometric methods in data science; formal proof verification in Lean 4; variational problems, geometric analysis, and mathematical physics; mathematical software development in Julia.

EDUCATION

Ph.D. in Mathematics, Texas Tech University, 2021–2025

B.Sc. (Hons.) Special in Mathematics, University of Peradeniya, Sri Lanka, 2014–2018

PROFESSIONAL EXPERIENCE

Assistant Professor, Tougaloo College, Spring 2025 – Present

Research / Teaching Assistant, Texas Tech University, 2021 – 2024

Lecturer, University of Moratuwa, Sri Lanka, 2019 – 2021

SELECTED PUBLICATIONS

- M. Samarakkody. “Formalizing the Classical Isoperimetric Inequality in the Two-Dimensional Case.” Submitted.
- Y. Li, E. Güler, M. Toda, M. Samarakkody. “Geometric Properties of Hypersurfaces of Right Conoids in Minkowski Spacetime.” Submitted.
- A. Pámpano, M. Samarakkody, H. Tran. “Closed p -Elastic Curves in Spheres of $\mathbb{L}^3$.” *Journal of Mathematical Analysis and Applications*, 542(2):129147, 2025.

GRANTS

- **PI**, NIH IDeA (P20GM103476), \$29,317 – Statistical/ML approaches for heart disease prediction.
- **PI**, SCALE Program (Purdue partnership), \$73,094.77 – Microelectronic workforce development for radiation-hardening.
- **Recipient**, NSF HBCU UP Implementation Project (Award No. 2510537).

SOFTWARE

- **IsoperimetricInequality** – Lean 4 formalization. DOI: 10.5281/zenodo.20258500
- **SymKit.jl** – Julia package for symbolic computation. DOI: 10.5281/zenodo.20294029
- **ParametricCurves.jl** – Julia package for parametric curves. DOI: 10.5281/zenodo.20288830

FELLOWSHIPS & AWARDS

Dr. Shelby Hildebrand Graduate Math Fellowship (\$10,000, 2023–2024); AT&T Graduate Fellowship (\$16,000, 2021–2025); University Prize for Academic Excellence, University of Peradeniya (2018).

TECHNICAL SKILLS

Languages: Python, Julia, C **Theorem Proving:** Lean 4 **Math Software:** Mathematica, MATLAB
Typesetting: LaTeX, Typst